Interactive Visualization Tool: Comparing Emissions and Costs for EVs and ICE Vehicles in Utah

How To Add Models

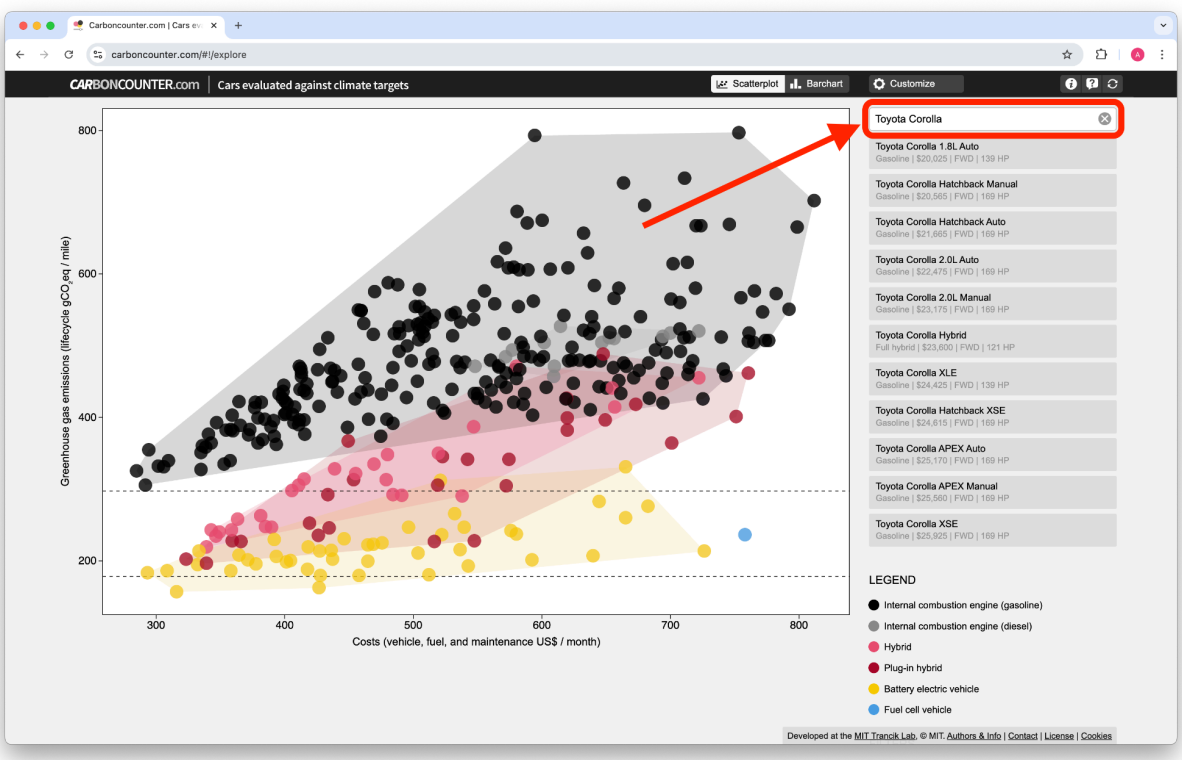
Summary:

Adding models to this tool requires manually pulling vehicle data from the [CarbonCounter tool](), as this is used as the source of truth for vehicle MSRP and manufacturing emissions. That data is manually input into the `vehiclestoinclude.xlsx` file and then joined with the EPA vehicles dataset by running the `create_vehicledata_file.ipynb`. That notebook creates the file, `vehicledata.csv`. Once this is created and updated, the Streamlit app should populate the new models into the web application.

Step By Step Instructions:

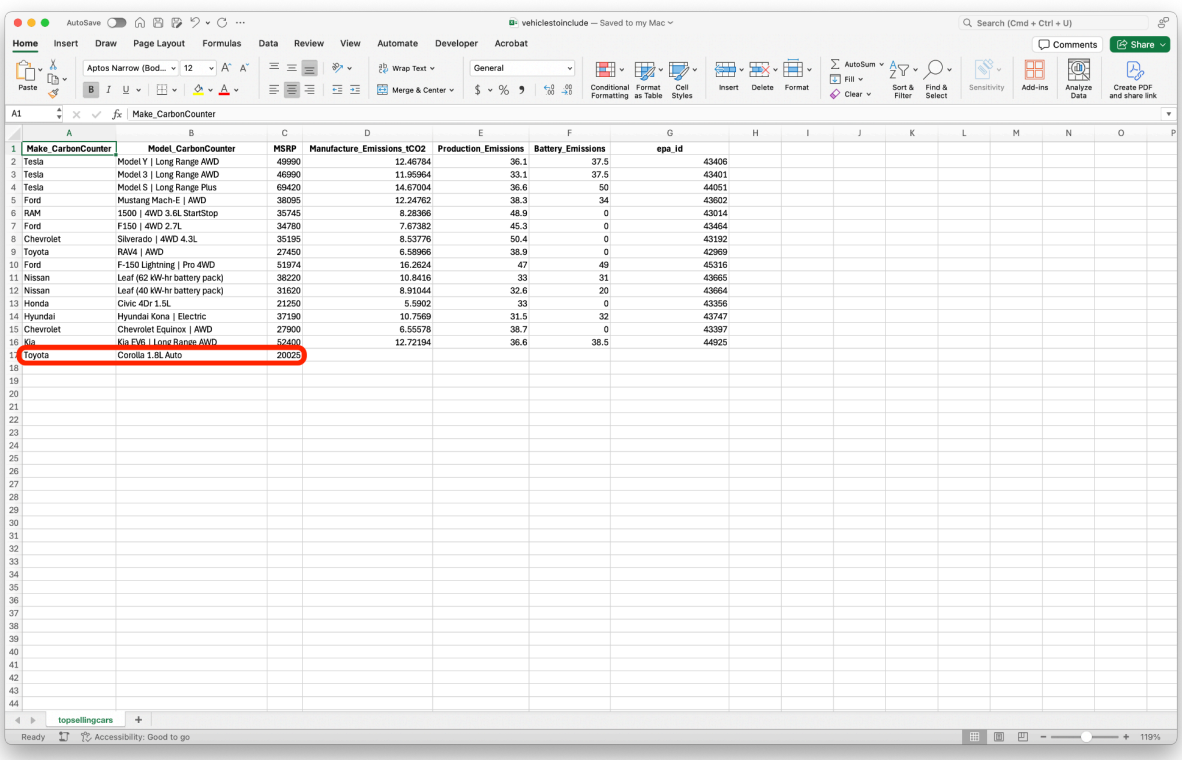
Step 1:

Find the vehicle to add in [Carbon Counter]() by searching in the bar in the top right of the tool and clicking on the desired option:



Step 2:

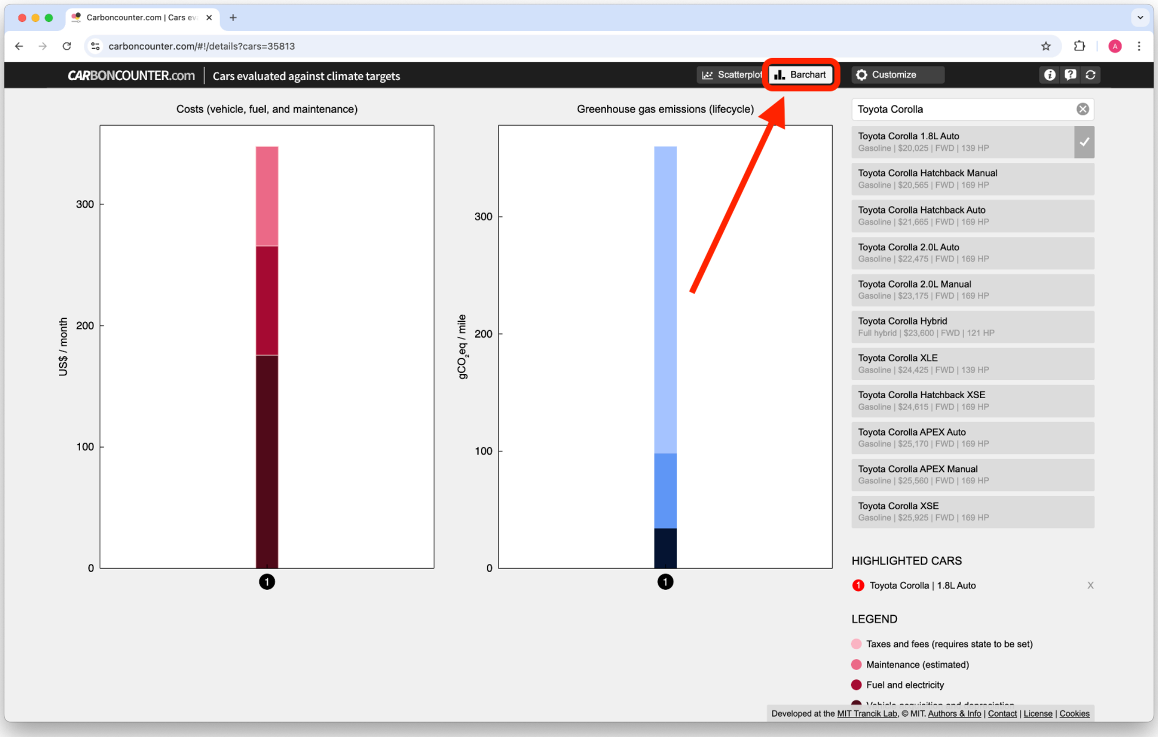
Open the `vehiclestoinclude.xlsx` file and copy and paste the Make, Model, and MSRP from your selected CarbonCounter model into the Excel file.



Make_CarbonCounter	Model_CarbonCounter	MSRP	Manufacture_Emissions_tCO2	Production_Emissions	Battery_Emissions	epa_id
Tesla	Model Y | Long Range AWD	49990	12.46784	36.1	37.5	43406
Tesla	Model 3 | Long Range AWD	46990	11.95964	33.1	37.5	43401
Tesla	Model S | Long Range Plus	69420	14.67004	36.6	50	44051
Ford	Mustang Mach-E | AWD	38095	12.24762	38.3	34	43602
RAM	1500 | 4WD 3.6L StartStop	35745	8.28366	48.9	0	43014
Ford	F150 | 4WD 2.7L	34780	7.67382	45.3	0	43464
Chevrolet	Silverado | 4WD 4.3L	35195	8.53776	50.4	0	43192
Toyota	RAV4 | AWD	27450	6.58966	38.9	0	42969
Ford	F-150 Lightning | Pro 4WD	51974	16.2624	47	49	45316
Nissan	Leaf (62 kW-hr battery pack)	38220	10.8416	33	31	43665
Nissan	Leaf (40 kW-hr battery pack)	31620	8.91044	32.6	20	43664
Honda	Civic 4Dr 1.5L	21250	5.5902	33	0	43356
Hyundai	Hyundai Kona | Electric	37190	10.7569	31.5	32	43747
Chevrolet	Chevrolet Equinox | AWD	27900	6.55578	38.7	0	43397
Kia	Kia EV6 | Long Range AWD	52400	12.72194	36.6	38.5	44925
Toyota	Corolla 1.8L Auto	20025				

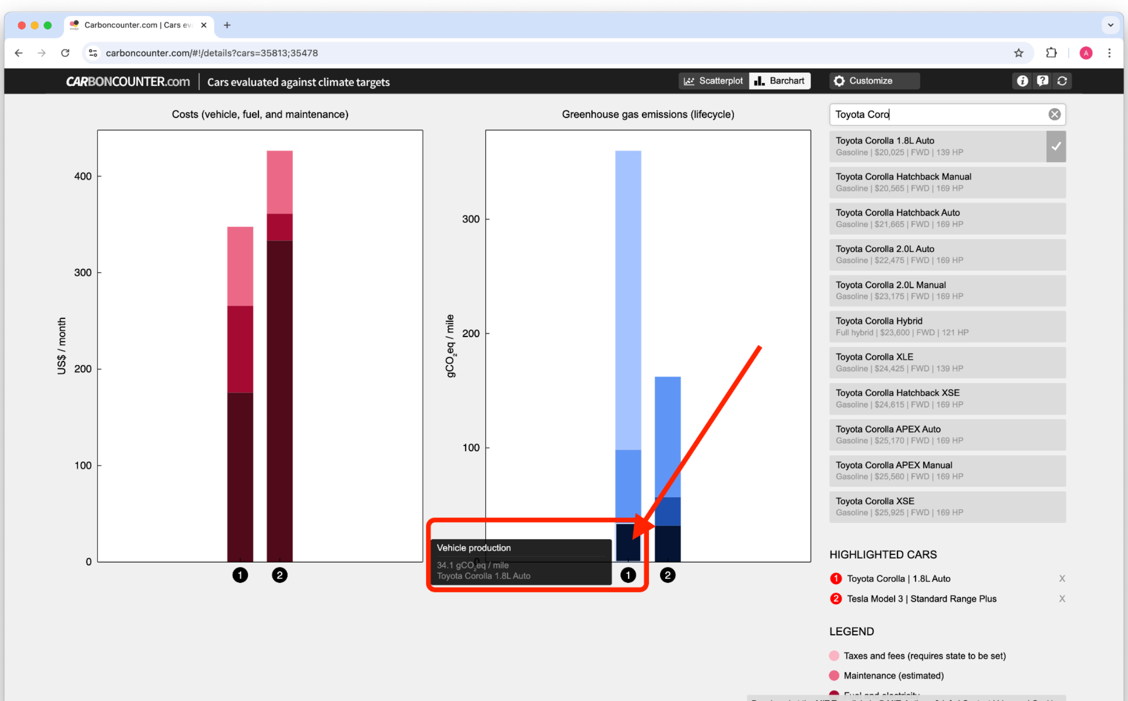
Step 3:

Click the “Barchart” button in the top right of the CarbonCounter tool.

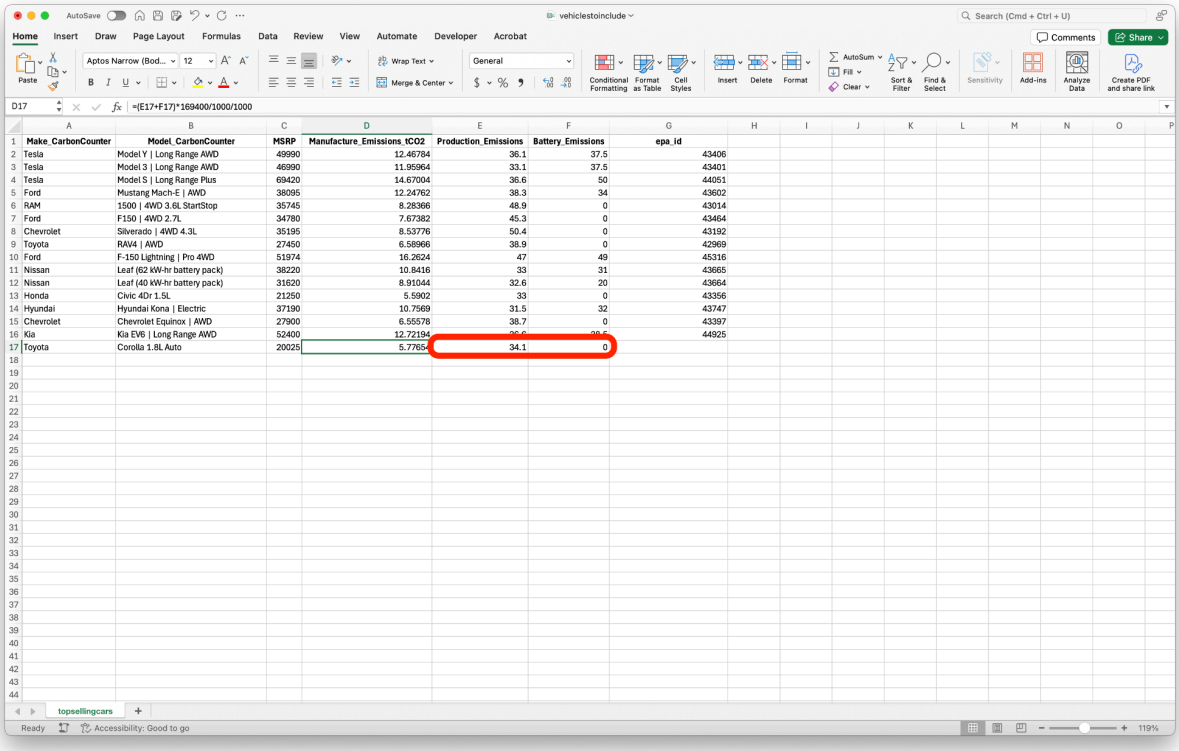


Step 4:

Transcribe the “Vehicle production” and (for EVs only) “Battery production” emissions metrics into the `vehiclestoinclude.xlsx` file by hovering over the dark blue and navy blue bar chart sections, respectively, in CarbonCounter.



This will automatically calculate the `Manufacture_Emissions_tCO2` column through the built-in formula.



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Chevrolet	Silverado | 4WD 4.3L	35195	8.53776	50.4	0	43192
Toyota	RAV4 | AWD	27450	6.58966	38.9	0	42969
Ford	F-150 Lightning | Pro 4WD	51974	16.2624	47	49	45316
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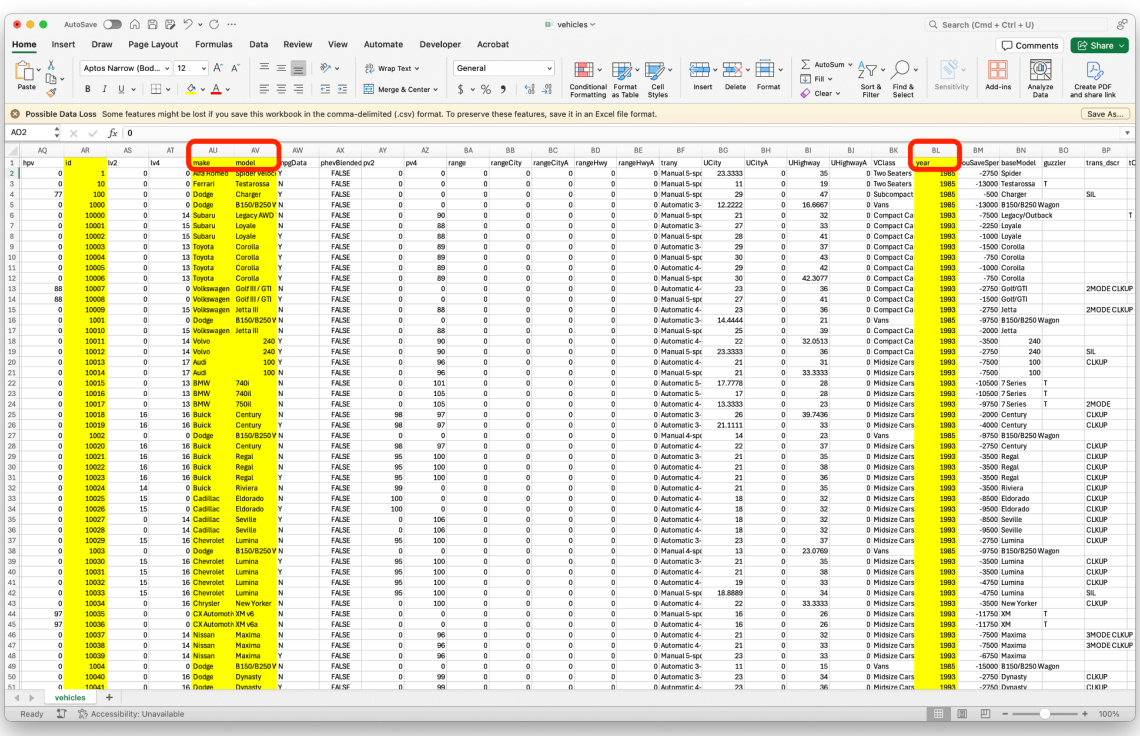
Chevrolet	Chevrolet Equinox | AWD	27900	6.55578	38.7	0	43397
Kia	Kia EV6 | Long Range AWD	52400	12.72194	[illegible]	[illegible]	44925
Toyota	Corolla 1.8L Auto	20025	5.77654	34.1	0	

Step 5:

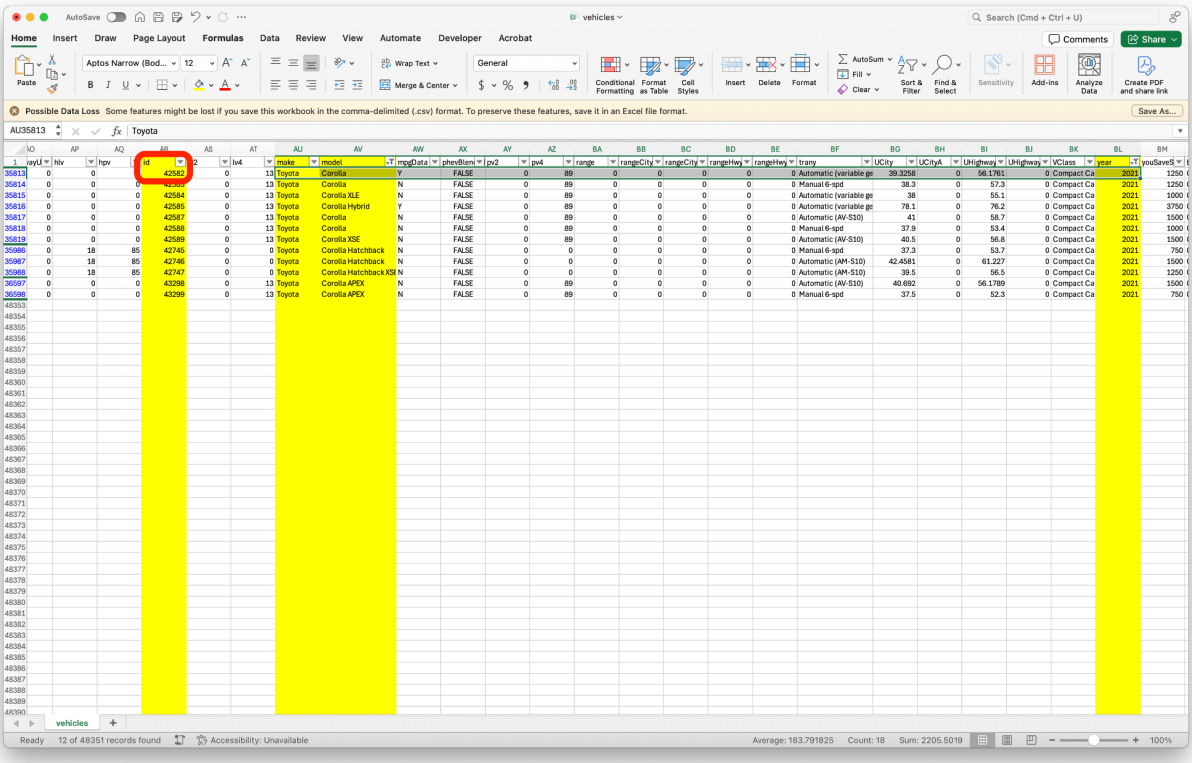
Find the same make/model/trim in the EPA's vehicles.csv by filtering the make (column AU), model (column AV), and year (column BL) columns.



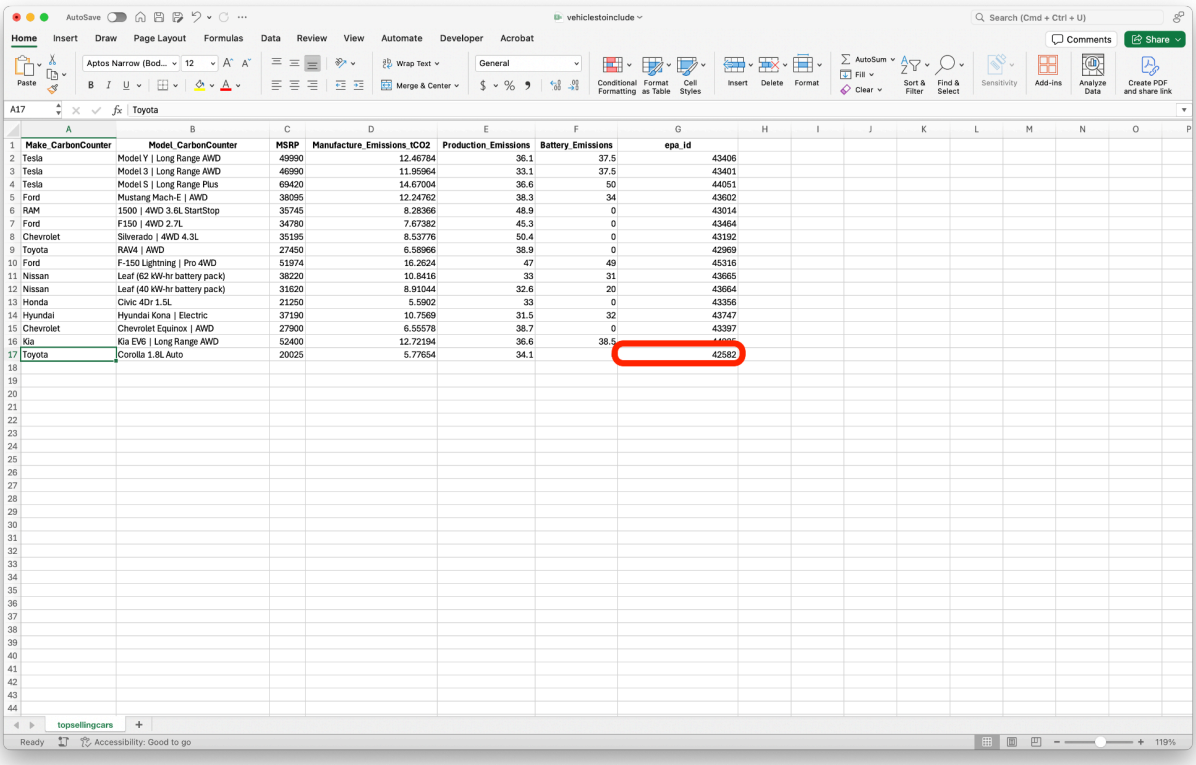
hpv	id	lv2	lv4	make	model	year
0	1	0	0	Alfa Romeo	Spider Veloci	1985
0	10	0	0	Ferrari	Testarossa	1985
77	100	0	0	Dodge	Charger	1985
0	1000	0	0	Dodge	B150/B250 V	1985
0	10000	0	14	Subaru	Legacy AWD	1993
0	10001	0	15	Subaru	Loyale	1993
0	10002	0	15	Subaru	Loyale	1993
0	10003	0	13	Toyota	Corolla	1993
0	10004	0	13	Toyota	Corolla	1993
0	10005	0	13	Toyota	Corolla	1993
0	10006	0	13	Toyota	Corolla	1993
88	10007	0	0	Volkswagen	Golf III / GTI	1993
88	10008	0	0	Volkswagen	Golf III / GTI	1993
0	10009	0	15	Volkswagen	Jetta III	1993
0	1001	0	0	Dodge	B150/B250 V	1985
0	10010	0	15	Volkswagen	Jetta III	1993
0	10011	0	14	Volvo	240	1993
0	10012	0	14	Volvo	240	1993
0	10013	0	17	Audi	100	1993
0	10014	0	17	Audi	100	1993
0	10015	0	13	BMW	740i	1993
0	10016	0	13	BMW	740il	1993
0	10017	0	13	BMW	750il	1993
0	10018	16	16	Buick	Century	1993
0	10019	16	16	Buick	Century	1993
0	1002	0	0	Dodge	B150/B250 V	1985
0	10020	16	16	Buick	Century	1993
0	10021	16	16	Buick	Regal	1993
0	10022	16	16	Buick	Regal	1993
0	10023	16	16	Buick	Regal	1993
0	10024	14	0	Buick	Riviera	1993
0	10025	15	0	Cadillac	Eldorado	1993
0	10026	15	0	Cadillac	Eldorado	1993
0	10027	0	14	Cadillac	Seville	1993
0	10028	0	14	Cadillac	Seville	1993
0	10029	15	16	Chevrolet	Lumina	1993
0	1003	0	0	Dodge	B150/B250 V	1985
0	10030	15	16	Chevrolet	Lumina	1993
0	10031	15	16	Chevrolet	Lumina	1993
0	10032	15	16	Chevrolet	Lumina	1993
0	10033	15	16	Chevrolet	Lumina	1993
0	10034	0	16	Chrysler	New Yorker	1993
97	10035	0	0	CX Automoti	XM v6	1993
97	10036	0	0	CX Automoti	XM v6a	1993
0	10037	0	14	Nissan	Maxima	1993
0	10038	0	14	Nissan	Maxima	1993
0	10039	0	14	Nissan	Maxima	1993
0	1004	0	0	Dodge	B150/B250 V	1985
0	10040	0	16	Dodge	Dynasty	1993
0	10041	0	16	Dodge	Dynasty	1993

Step 6:

For the row you've selected, copy and paste the `id` column (column AR) into the `epa_id` column (column G) in the `vehiclestoinclude.xlsx` file.



	AO	AP	AQ	AR	AS	AT	AU	AV	AW	AX	AY	AZ	BA	BB	BC	BD	BE	BF	BG	BH	BI	BJ	BK	BL	BM
1	ayU	hlv	hpv	id	2	tv4	make	model	mpgData	phevBlen	pv2	pv4	range	rangeCity	rangeCity	rangeHwy	rangeHwy	trany	UCity	UCityA	UHighway	UHighway	VClass	year	youSaveS
35813	0	0		42582	0	13	Toyota	Corolla	Y	FALSE	0	89	0	0	0	0	0	Automatic (variable ge	39.3258	0	56.1761	0	Compact Ca	2021	1250
35814	0	0	0	42583	0	13	Toyota	Corolla	N	FALSE	0	89	0	0	0	0	0	Manual 6-spd	38.3	0	57.3	0	Compact Ca	2021	1250
35815	0	0	0	42584	0	13	Toyota	Corolla XLE	N	FALSE	0	89	0	0	0	0	0	Automatic (variable ge	38	0	55.1	0	Compact Ca	2021	1000
35816	0	0	0	42585	0	13	Toyota	Corolla Hybrid	Y	FALSE	0	89	0	0	0	0	0	Automatic (variable ge	78.1	0	76.2	0	Compact Ca	2021	3750
35817	0	0	0	42587	0	13	Toyota	Corolla	N	FALSE	0	89	0	0	0	0	0	Automatic (AV-S10)	41	0	58.7	0	Compact Ca	2021	1500
35818	0	0	0	42588	0	13	Toyota	Corolla	N	FALSE	0	89	0	0	0	0	0	Manual 6-spd	37.9	0	53.4	0	Compact Ca	2021	1000
35819	0	0	0	42589	0	13	Toyota	Corolla XSE	N	FALSE	0	89	0	0	0	0	0	Automatic (AV-S10)	40.5	0	56.8	0	Compact Ca	2021	1500
35986	0	18	85	42745	0	0	Toyota	Corolla Hatchback	N	FALSE	0	0	0	0	0	0	0	Manual 6-spd	37.3	0	53.7	0	Compact Ca	2021	750
35987	0	18	85	42746	0	0	Toyota	Corolla Hatchback	N	FALSE	0	0	0	0	0	0	0	Automatic (AM-S10)	42.4581	0	61.227	0	Compact Ca	2021	1500
35988	0	18	85	42747	0	0	Toyota	Corolla Hatchback XSE	N	FALSE	0	0	0	0	0	0	0	Automatic (AM-S10)	39.5	0	56.5	0	Compact Ca	2021	1250
36597	0	0	0	43298	0	13	Toyota	Corolla APEX	N	FALSE	0	89	0	0	0	0	0	Automatic (AV-S10)	40.692	0	56.1789	0	Compact Ca	2021	1500
36598	0	0	0	43299	0	13	Toyota	Corolla APEX	N	FALSE	0	89	0	0	0	0	0	Manual 6-spd	37.5	0	52.3	0	Compact Ca	2021	750



	A	B	C	D	E	F	G
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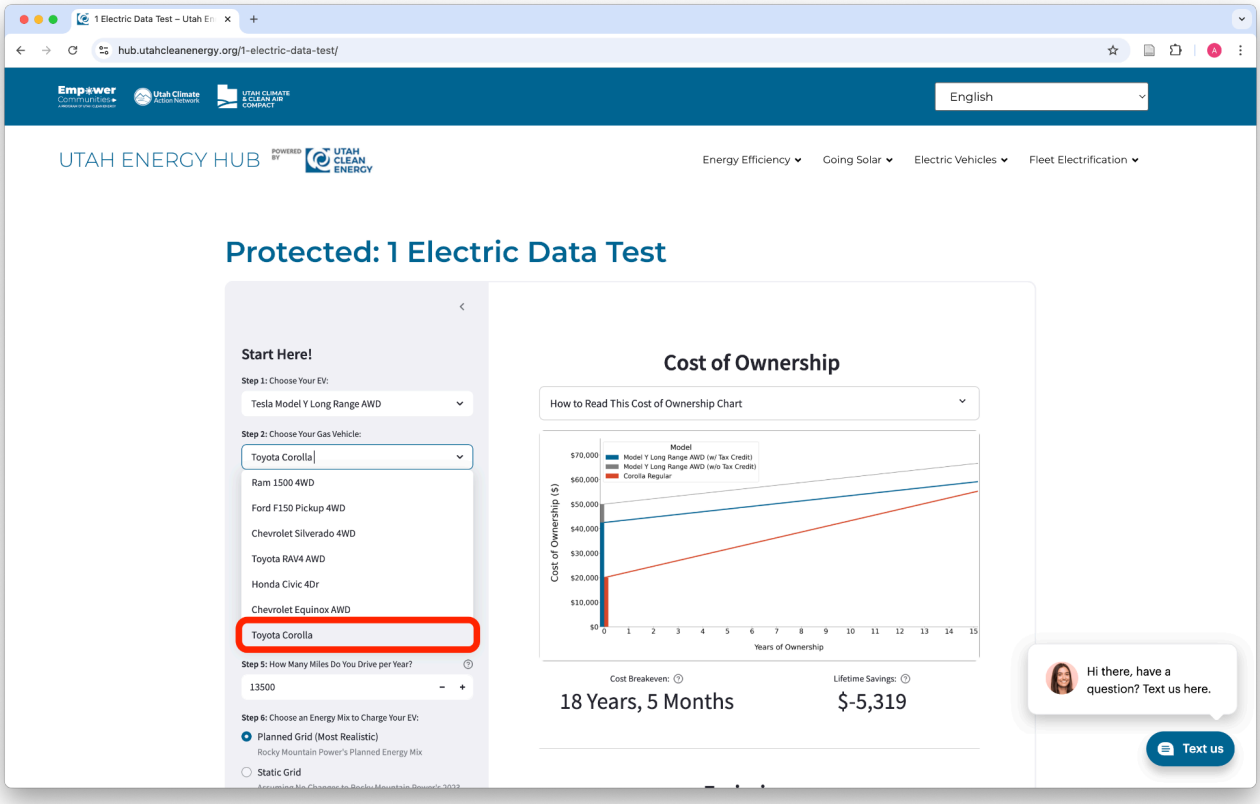
Step 7:

Run the `create_vehicledata_file.ipynb` to create the `vehicledata.csv` file.

Step 8:

All done! Your new model will now populate in the web application!